Python Operators Complete Notes

Introduction to Operators

Operators are special symbols used to perform operations on variables and values.

Example:

```
x = 12
y = 4

print(x + y)
```

Output:

```
16
```

Types of Operators in Python

1. Arithmetic Operators
2. Assignment Operators
3. Comparison Operators
4. Logical Operators
5. Identity Operators
6. Membership Operators
7. Bitwise Operators

1. Arithmetic Operators

Arithmetic operators are used to perform mathematical calculations.

Operator	Meaning
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Modulus
//	Floor Division
**	Exponent

Addition Operator (+)

Adds two values.

```
price1 = 250
price2 = 150

total = price1 + price2

print(total)
```

Output:

```
400
```

Subtraction Operator (-)

Subtracts one value from another.

```
total_marks = 950
obtained_marks = 875

remaining = total_marks - obtained_marks

print(remaining)
```

Output:

```
75
```

Multiplication Operator (*)

Multiplies values.

```
ticket_price = 120
tickets = 5

bill = ticket_price * tickets

print(bill)
```

Output:

```
600
```

Division Operator (/)

Returns quotient in decimal format.

```
distance = 450
days = 6

average = distance / days

print(average)
```

Output:

```
75.0
```

Modulus Operator (%)

Returns remainder.

```
students = 53
groups = 5

print(students % groups)
```

Output:

```
3
```

Floor Division Operator (//)

Returns only integer quotient.

```
candies = 29
children = 4

print(candies // children)
```

Output:

```
7
```

Exponent Operator (**)

Calculates power.

```
print(3 ** 4)
```

Output:

```
81
```

2. Assignment Operators

Assignment operators are used to assign values.

Operator	Example
=	x = 5
+=	x += 5
-=	x -= 5
*=	x *= 5
/=	x /= 5
%=	x %= 5
//=	x //= 5
**=	x **= 5

Equal Operator (=)

```
salary = 45000

print(salary)
```

Output:

```
45000
```

Plus Equal Operator (+=)

```
wallet = 200
wallet += 50

print(wallet)
```

Output:

```
250
```

Minus Equal Operator (-=)

```
battery = 100
battery -= 30

print(battery)
```

Output:

```
70
```

Multiply Equal Operator (*=)

```
points = 8
points *= 3

print(points)
```

Output:

```
24
```

Divide Equal Operator (/=)

```
amount = 900
amount /= 3

print(amount)
```

Output:

```
300.0
```

Modulus Equal Operator (%=)

```
number = 47
number %= 6

print(number)
```

Output:

5

Floor Division Equal Operator (//=)

```
money = 1250
money //= 12

print(money)
```

Output:

```
104
```

Exponent Equal Operator (**=)

```
value = 5
value **= 2

print(value)
```

Output:

```
25
```

3. Comparison Operators

Comparison operators compare values and return True or False.

Operator	Meaning
==	Equal To
!=	Not Equal To
>	Greater Than
<	Less Than
>=	Greater Than or Equal To
<=	Less Than or Equal To

Equal To Operator (==)

```
pin1 = 1234
pin2 = 1234

print(pin1 == pin2)
```

Output:

True

Not Equal To Operator (!=)

```
city1 = "Delhi"
city2 = "Mumbai"

print(city1 != city2)
```

Output:

True

Greater Than Operator (>)

```
temperature_today = 38
temperature_yesterday = 34

print(temperature_today > temperature_yesterday)
```

Output:

True

Less Than Operator (<)

```
age_rahul = 19
age_arjun = 24

print(age_rahul < age_arjun)
```

Output:

True

Greater Than or Equal To (>=)

```
required_marks = 35
student_marks = 35

print(student_marks >= required_marks)
```

Output:

True

Less Than or Equal To (<=)

```
speed_limit = 80
car_speed = 75

print(car_speed <= speed_limit)
```

Output:

```
True
```

4. Logical Operators

Logical operators combine conditions.

Operator	Meaning
and	True if both conditions are True
or	True if at least one condition is True
not	Reverses result

AND Operator

```
age = 22
has_license = True

print(age >= 18 and has_license)
```

Output:

```
True
```

OR Operator

```
username = "admin"
email = "admin@gmail.com"

print(username == "admin" or email == "admin@gmail.com")
```

Output:

```
True
```

NOT Operator

```
is_raining = False

print(not is_raining)
```

Output:

```
True
```

5. Identity Operators

Identity operators compare memory locations.

Operator	Meaning
is	Same object
is not	Different object

IS Operator

```
list_a = ["apple", "banana"]
list_b = list_a

print(list_a is list_b)
```

Output:

```
True
```

IS NOT Operator

```
x = [10, 20]
y = [10, 20]

print(x is not y)
```

Output:

```
True
```

6. Membership Operators

Membership operators check whether a value exists in a sequence.

Operator	Meaning
in	Present
not in	Not Present

IN Operator

```
languages = ["Python", "Java", "C++"]

print("Python" in languages)
```

Output:

```
True
```

NOT IN Operator

```
colors = ["Red", "Blue", "Green"]

print("Black" not in colors)
```

Output:

```
True
```

7. Bitwise Operators

Bitwise operators work on binary numbers.

Operator	Meaning
&	AND
^	XOR
~	NOT
<<	Left Shift
>>	Right Shift

Bitwise AND (&)

```
a = 12
b = 10

print(a & b)
```

Output:

```
8
```

Explanation:

```
12 = 1100
10 = 1010
--------------
    1000 = 8
```

Bitwise OR (|)

```
a = 12
b = 10

print(a | b)
```

Output:

```
14
```

Bitwise XOR (^)

```
a = 12
b = 10

print(a ^ b)
```

Output:

```
6
```

Bitwise NOT (~)

```
num = 7

print(~num)
```

Output:

-8

Left Shift Operator (<<)

```
number = 6

print(number << 2)
```

Output:

24

Right Shift Operator (>>)

```
number = 20

print(number >> 2)
```

Output:

5

Operator Precedence

Operator precedence decides execution order.

Priority	Operators
Highest	()
	**
	*, /, //, %
	+, -
	Comparison Operators
	not
	and
Lowest	or

Example

```
result = 18 - 4 * 2

print(result)
```

Output:

```
10
```

Explanation:

```
4 * 2 = 8
18 - 8 = 10
```

Difference Between == and is

==	is
Compares values	Compares memory location
Checks equality	Checks object identity

Example

```
a = [1, 2]
b = [1, 2]

print(a == b)
print(a is b)
```

Output:

```
True
False
```

Important Interview Questions

1. What is an operator in Python?
2. Difference between / and //
3. Difference between == and is
4. What is operator precedence?
5. Explain logical operators with examples.
6. What are membership operators?
7. Explain bitwise operators.
8. What does modulus operator do?
9. Which operators return Boolean values?
10. Explain assignment operators with examples.

Quick Revision

Arithmetic Operators

+ - * / % // **

Assignment Operators

= += -= *= /= %= //= **=

Comparison Operators

== != > < >= <=

Logical Operators

and or not

Identity Operators

is is not

Membership Operators

in not in

Bitwise Operators

& | ^ ~ << >>